Local Income Taxes and Residential Sorting:

Evidence from Pennsylvania LEHD Origin-Destination Data

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Abstract

Do local income tax rate changes cause residential sorting across municipal boundaries? Using LEHD Origin-Destination data linking each worker's residence block to their workplace block at annual frequency, I estimate the effect of Pennsylvania municipal Earned Income Tax (EIT) rate changes on residential and workplace outcomes for ∼2,500 cousubs over 2002–2019. The PA EIT bill is piecewise: $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where τ^{home} stacks the muni resident rate and the school district rate. A rate hike only *moves the bill* when the relevant side of the max binds. I exploit this structure with a unified muni-level design that decomposes each rate-change event by direction (Inc/Dec) and binding-state transition (already-binding, becomes-binding, non-binding-throughout), running five staggered-DiD estimators on a single panel with three treatments (muni resident rate, school district rate, muni nonresident rate). The within-design placebo is the Inc-nobind cell—rate hikes whose post-event home stack remains below the nonresident rate, so the bill never moves. Using stacked continuous DiD, a 1 percentage-point muni resident rate hike in the binding cell reduces `live_and_work` workers by 40 percent, total jobs by 27 percent, and total residents by 10 percent; the share of residents commuting out rises 4.1 percentage points. The

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Inc-nobind placebo cell is null on every level outcome. School district rate changes do not produce sorting at any binding state, even when SD changes flip the home-stack binding state. The work-side nonresident rate moves nonresident workers but not residents—channel separation is clean. Industry-decomposed outcomes (LODES WAC) show consumer-facing jobs (retail, food, arts, other services) fall 30 percent in the binding cell while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated, consistent with a local-services demand spillover from residents leaving rather than direct nonresident-tax incidence. The pooled-cell pre-trend test is null; city-subset pre-trends are present, consistent with selection into rate hikes by post-industrial cities and limiting causal interpretation of the city-specific magnitude.

1 Introduction

The Tiebout hypothesis—that households "vote with their feet" by sorting across jurisdictions offering different bundles of taxes and public services (Tiebout, 1956)—remains a central organizing principle in local public finance. A large empirical literature documents tax-driven migration at the state level for high earners (Kleven et al., 2020; Young et al., 2016; **?**) and at sub-national borders for sales taxes (Agrawal, 2015). Yet sub-state income-tax sorting—the canonical Tiebout margin, where moving costs are small and tax boundaries are crossed by a short relocation—is essentially untested in the United States. Two recent studies of Indiana and Maryland local income taxes (**??**) use county-level IRS data and find county-level migration responses but cannot resolve the muni-level variation that defines the actual tax boundary. A literature survey by **?** explicitly identifies sub-state local-income-tax sorting as an open question, and **?** characterize the empirical evidence as "thin."

The gap persists for two reasons. First, the jurisdictions that levy local income taxes are sub-county, so standard migration data sources do not see them. The IRS Statistics of Income tracks county-to-county migration; the American Community Survey measures place-level population but does not link workers to employers. Second, local income-tax systems feature credit mechanisms, overlapping school-district taxes, and uneven enforcement that complicate the mapping from statutory rates to effective tax burden. A naive comparison of rates and population changes conflates tax incidence with amenity valuation, fiscal capacity, and enforcement quality.

This paper addresses both challenges using LEHD Origin-Destination Employment Statistics (LODES), the only administrative data source that links each worker's residence tract to their workplace tract at annual frequency. The setting is Pennsylvania, 2002–2019. Pennsylvania operates a universal Earned Income Tax under Act 511 of 1965 with continuous rate variation across approximately 2,500 municipalities and 500 school districts, and a quasi-experimental enforcement break: Act 32 of 2008 (effective January 1, 2012) consolidated EIT collection into 69 Tax Collection Districts and mandated employer withholding at the

higher of the home resident rate or the work nonresident rate. Before Act 32, withholding was fragmented and many municipalities did not even publish a nonresident rate. The structural break in enforcement—same statutory rates, very different effective tax burden—anchors causal identification.

This paper makes three contributions. First, it identifies the binding structure of the PA EIT as the central margin for empirical identification. The bill is piecewise in $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, so a rate change only moves the bill when its side binds. I classify each muni-rate event by direction (Inc/Dec) and binding-state transition relative to the muni's nonresident rate, yielding seven cells (six event cells + NC controls). The within-design placebo is the Inc-nobind cell: rate hikes whose post-event home stack stays below the nonresident rate, so the bill never moves and theory predicts no behavioral response.

Second, the empirical design is a unified muni-level panel with three simultaneous treatments—muni resident rate, school district rate, and muni nonresident rate, each separately decomposed by binding cells—and five staggered-DiD estimators run in parallel. The headline estimator is stacked continuous DiD; Q2 TWFE pooled, single-event TWFE, stacked binary, and DCDH (de Chaisemartin et al., 2024) provide triangulation. Controls are share-based throughout: police, streets, and debt service as shares of total expenditures; EIT, real estate, and intergovernmental revenue as shares of total revenues. The never-changed control pool is strictly defined as munis whose rate never changed across the panel, not the broader set of munis whose first event could be cell-classified.

Third, the headline finding is that the muni resident rate moves residents through the binding channel only. A 1 pp muni-rate hike in the binding cell reduces `live_and_work` workers by 40 percent, total jobs by 27 percent, total residents by 10 percent, and raises the share of residents commuting out by 4.1 percentage points. The Inc-nobind placebo cell—rate hikes whose post-event home stack stays below the nonresident rate—is null on every level outcome. School district rate changes do not produce sorting in any cell, even when SD changes flip the binding state, consistent with workers responding to the bill rather than

to the rate's identity. The nonresident rate moves in-commuters but not residents. Industry decomposition using LODES WAC shows consumer-facing jobs fall 30 percent in the binding cell while tradeable jobs are insulated, consistent with local-services demand spillover from residents leaving rather than direct nonresident-tax incidence.

The headline magnitude is concentrated in 8 post-industrial PA cities (Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell). The 32 borough and township Inc-bind munis show direction-consistent but individually insignificant effects. Pre-trend diagnostics on the pooled Inc-bind cell are null in the binary Sun-Abraham event study; pre-trends on the city subset alone are significant, indicating selection into rate hikes by cities already on declining trajectories. I therefore present the city-specific magnitude as "rate hikes in declining cities are associated with continued decline" rather than as a clean causal estimate, and report the pooled headline as the defensible quantity.

2 Institutional Background

Pennsylvania's Earned Income Tax (EIT) is the largest source of local-income-tax variation in the United States. Approximately 2,500 municipalities and 500 school districts each set their own rate, covering essentially all earned income earned in or by Pennsylvania residents. Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries within the same metropolitan area. This creates sharp spatial discontinuities in tax burden that can influence where workers choose to live and work.

2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act (Act 511 of 1965). It applies to all earned income with no minimum threshold; retirees are exempt. The standard combined cap is 1%, typically split as 0.5% municipal and 0.5%

school district. Home Rule municipalities may exceed this cap, and Philadelphia operates separately under the Sterling Act of 1932 at approximately 3.9%. The EIT is the primary revenue lever for most Pennsylvania municipalities: property tax millage caps are binding in many jurisdictions. Our panel (2002–2019) contains 210 muni rate increases, 119 muni rate decreases, 611 school-district rate increases, and 52 school-district rate decreases.

Three rates per municipality. Each Pennsylvania municipality has three EIT rates relevant to a worker:

- $\tau^{\text{res,muni}}$, the muni resident rate, levied on residents of the muni;
- $\tau^{\text{nr,muni}}$, the muni nonresident rate, levied on workers employed in the muni who live elsewhere;
- τ^{SD}, the school-district rate, levied on residents only (school districts cannot tax nonresidents).

Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (muni, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

The piecewise tax allocation. Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where j is the worker's home muni and i is the work muni. The work jurisdiction always collects its full nonresident rate first; the home muni and school district then split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate

exceeds the home stack, the home jurisdictions get nothing. The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\big(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\big). \tag{2}$$

Implications for sorting. Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate muni or school district. When the work nonresident rate binds, the worker can lower the bill only by changing job to a lower-nonres-rate work location; moving home without changing job has no effect. For workers who live and work in the same muni, only the home stack applies—they pay $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$.

A worker who lives in muni j and moves to a lower-rate muni j' while keeping the same job at i saves at least the SD rate at j: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD}. The often-stated claim that "moving residence under a credit system yields zero tax savings" is wrong under Pennsylvania's piecewise structure—residence moves do save tax, because the home-side stack includes the SD rate that does not reappear at the work side.

Philadelphia (excluded from the Act 32 framework). Philadelphia operates under the Sterling Act of 1932 rather than Act 32. Philadelphia residents pay the city resident rate regardless of work location with no offset; Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania munis. Our main estimates retain Philadelphia in the sample for completeness; results are robust to dropping it.

Act 32: Tax Collection Consolidation. Before 2012, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers,

and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The piecewise rule described in Equation 1 existed on paper but was not effectively enforced.

Act 32 of 2008 (effective January 1, 2012) transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *binding*: rate changes now mechanically appear in every worker's paycheck through employer withholding, regardless of whether they reside in the taxing municipality. Second, it eliminated the possibility of de facto tax avoidance through residential relocation. Before Act 32, moving away from a rate-raising city could effectively reduce a worker's tax burden because the work municipality lacked the enforcement capacity to collect from nonresidents. After Act 32, the credit system is fully enforced, and moving away while keeping the same job provides no tax relief.

School District EIT. School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across our panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework: Binding-State Decomposition

The piecewise tax allocation in Equation (1) is the analytical engine of the empirical design. A worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side of the max binds. This generates a sharp prediction: the same statutory rate change can produce a strong behavioral response, a moderate response, or no response at all depending on the binding state pre- and post-event.

3.1 Six Cells per Side

For each muni i and each rate variable $\tau \in \{\tau^{\text{res,muni}}, \tau^{\text{SD}}, \tau^{\text{nr,muni}}\}$, classify the muni's first observed rate event by direction (Increase / Decrease) and binding-state transition relative to the muni's own home-stack vs nonresident rate comparison:

- **Inc-bind**: rate up; home stack was binding pre and post. Bill rises by full $\Delta\tau$ for in-muni residents. *Predicted: strongest behavioral response.*
- **Inc-becomes**: rate up; home stack flips from non-binding to binding. Bill rises by less than full $\Delta\tau$ (only the portion above the work nonres rate). *Predicted: intermediate response.*
- **Inc-nobind**: rate up; home stack stays below nonres rate throughout. *Bill never moves. Predicted: no response. This is the within-design placebo.*
- **Dec-bind**: rate down; home stack stays binding. Bill falls by full $\Delta\tau$. Predicted mirror of Inc-bind under symmetry.
- **Dec-becomes-non**: rate down; home stack flips from binding to non-binding. Bill falls only down to nonres rate.

- **Dec-nobind**: rate down; home stack stays below nonres throughout. Bill never moves. Second placebo cell.

A seventh category, NC (*never-changed*), comprises munis whose τ never changes across the panel. NC serves as the control pool.

The fundamental testable prediction of the institutional rule is that **Inc-bind effects > Inc-becomes effects > Inc-nobind effects ≈ 0 in absolute value**, with Inc-nobind serving as the within-design placebo. If the Inc-nobind cell shows behavioral response, the design's mechanism interpretation fails—the response is something other than the binding rule.

3.2 Three Treatment Margins

PA's institutional structure separates three rate variables that change independently across muni-years:

- $\tau^{\text{res,muni}}$ (muni resident rate) directly affects the bill of in-muni residents through the home-stack channel.
- τ^{SD} (school district rate) also affects the home stack but is set by school districts, not municipalities. Predicted: same binding-rule mechanism as muni rate, but the SD's institutional decision-making is different from the muni's.
- $\tau^{\text{nr,muni}}$ (muni nonresident rate) affects nonresident workers commuting in. The piecewise rule means it moves the bill of workers from low-home-stack origins.

The two predictions across treatments:

1. Each rate's binding-cell decomposition shows the same Inc-bind > Inc-becomes > Inc-nobind pattern.

2. Channel separation: muni resident rate moves residents (live-and-work, total residents); muni nonresident rate moves nonresident workers (in-commuters); SD rate's effect depends on whether SD changes induce similar residential responses.

3.3 What This Replaces in the Prior Literature

The standard local-tax-incidence design pools all rate changes within a muni and estimates a single elasticity. This implicitly averages across the binding cells: an Inc-nobind muni-rate hike (no bill change) gets the same coefficient weight as an Inc-bind hike (full bill change), attenuating the average. The unified design separates them and generates a within-design placebo (Inc-nobind) that the pooled spec cannot.

The standard pre-/post-Act 32 difference-in-differences design compares the same muni's behavior before and after enforcement consolidation. This is informative about *when* sorting occurs but cannot distinguish enforcement from other 2012 shocks. The binding-state decomposition is post-2012 in our preferred sample (cell classification uses Act-32-era nonres rates) but identifies the response from the *cross-section* of muni binding states, not the time series. Both designs can be run; we report the unified design as the headline.

4 Data

4.1 Data Construction Pipeline

The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The analysis panels are constructed through the following steps:

1. **Scrape PA DCED MunStats tax rates.** PA does not publish a downloadable panel of historical EIT rates. The only source is the DCED MunStats portal, which serves one XLS file (∼12 MB) per year via an ASP.NET form. A Python script submits

requests for each year 2000–2026, extracting resident and nonresident EIT rates, school district EIT rates, and Local Services Tax for all ∼2,500 municipalities. About 27% of munis report multiple school-district rates because their territory spans multiple school districts; in the muni-level panel we assign the maximum SD rate per muni-year.

2. **Scrape PA DCED Annual Financial Reports.** Fiscal controls and revenue shares are constructed from the DCED Annual Financial Reports (71 columns per municipality-year) for 2002–2023, trimmed at load to 2002–2019. Variables include log spending on police, streets, debt service, and intergovernmental state aid, plus revenue shares (EIT, real estate tax, intergov as fractions of total revenue).

3. **Backfill nonresident rates.** The DCED nonresident registry was incomplete pre-Act 32: only 55–60% of munis reported a nonresident rate in 2002–2011. Coverage jumped to 72% in 2012 (Act 32 effective date) and plateaued near 77% by 2019. Diagnostics show this is a reporting gap, not a regime gap: only 0.2% of late-reporting munis have a true zero nonresident rate, and 96.3% of nonresident rates are perfectly stable across 2012–2019. We backfill missing pre-2012 nonresident rates with the earliest observed post-2012 rate per muni and flag the imputation.

4. **Download LODES origin-destination files at the block level.** LODES Version 8 (2020 Census geography throughout, retroactively applied to 2002–2021 by LEHD) is downloaded at the Census block level (`agg_geo = "block"` in `lehdr`). We download main files (intra-state) and auxiliary files (cross-state, where the workplace is in-state but the residence is out-of-state) for PA, 2002–2021. Auxiliary files are necessary because excluding cross-state commuters would mechanically inflate share_live_in for border munis. Segments include all jobs (S000), earnings categories (SE01, SE03), and age groups (SA01–SA03). Reference date: April 1 of each year.

5. **Crosswalk PA DCED to Census geography.** PA rates use DCED municipality IDs with no Census equivalent. PA municipalities are county subdivisions (COUSUBFP),

not Census places. We match DCED municipality names to TIGER/Line COUSUBFP codes using normalized name + county. Match rate: 99.6%. The 9 unmatched munis are boroughs and townships merged or dissolved before the 2020 Census.

6. **Build a block-to-cousub crosswalk.** LODES geographies (block GEOIDs) do not directly identify the cousub. We construct a block-to-cousub crosswalk from the TIGER 2020 PA blocks shapefile by point-in-polygon (block centroid `st_within` cousub). Because 2020 Census blocks nest within cousubs by construction, this assignment is exact—every block has an unambiguous cousub. The crosswalk recovers all 2,561 PA cousubs. We adopt block-level aggregation rather than the tract-level alternative because in rural PA a single tract often spans multiple cousubs; assigning each tract to one cousub by tract centroid silently drops 1,082 of 2,561 cousubs (42%) and misattributes employment for the cousubs it does retain.

7. **Compute neighbor pairs.** Centroid-to-centroid distances between all PA muni pairs; retain pairs within 80 km (the plausible commuting and relocation zone).

8. **Aggregate block-level LODES to municipality-level OD panels.** Raw block-level LODES has $\sim$5 million block-pair-year observations for PA per year. Each block-pair is mapped to a (work cousub, home cousub) pair via the block-to-cousub crosswalk and summed to produce, for each (work muni, year): total jobs, live-and-work (workers who reside in the work muni), commuters from other in-state munis, and cross-state commuters. This produces `share_live_in`, the primary outcome: the fraction of a city's workforce that also lives there.

9. **Merge rates and controls.** Final step joins rates and fiscal controls to the OD panel, producing the analysis-ready municipality $\times$ year dataset.

4.2 Data Sources

LODES Origin-Destination Data. LEHD LODES Version 8, 2002–2019. Block-level residence–workplace links derived from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages (**?**). We trim 2020–2021 due to COVID-related disruptions in both employment patterns and LODES coverage. The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality; the mechanism is calibrated so that aggregation to larger geographies produces estimates that approximate ground truth and have been validated against external sources at the tract level and above (**???**). We aggregate blocks to county subdivisions (cousubs) using the TIGER 2020 block-to-cousub crosswalk described in Section 4.1, item 3.

Municipal Tax Rate Panel. Pennsylvania: DCED MunStats, scraped annually. Resident and nonresident municipal EIT rates, school district EIT rates for all ∼2,500 municipalities. Name-based crosswalk to Census COUSUBFP (99.6% match rate). Pre-2012 nonresident rates backfilled from earliest observed post-2012 value per muni (the registry was incomplete pre-Act 32).

Municipal Financial Reports. PA DCED Annual Financial Reports, 2002–2019 (trimmed from 2002–2023 to exclude COVID/ARPA fiscal distortion). 71 columns per municipality-year including EIT revenue, real estate tax revenue, police expenditure, streets expenditure, total debt, state intergovernmental aid, population, and assessed market value of real estate.

Crime. FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest (Jacob Kaplan concatenated files), 2002–2019. Monthly agency-level counts of Part I index crimes (murder, robbery, aggravated assault, burglary, larceny-theft, motor vehicle theft). Aggregated to annual totals per agency. Matched to PA municipalities via `fips_place_code` (93.5% agency match rate).

Building Permits. Census Bureau Building Permits Survey (BPS), place-level annual files, 2002–2019. Total housing units permitted by structure type (1-unit, 2-unit, 3–4 unit, 5+ unit). PA municipalities matched via `fips_mcd_code` = COUSUBFP (93.3% MCD match rate).

4.3 Sample Restrictions and Transformations

1. **Sample period: 2002–2019.** LODES coverage begins in 2002. We trim at 2019 because COVID disrupted both commuting patterns (LODES 2020–2021) and municipal finances (ARPA transfers distort AFR data 2020–2023). Including these years would conflate pandemic-driven relocation with tax-motivated sorting.

2. **Minimum employment threshold: 50 total jobs.** Small municipalities have share_live_in values dominated by sampling noise (a single worker moving changes the ratio by $1/N$). The 50-job threshold balances precision against sample size. Sensitivity analysis (Section 7) confirms stability across thresholds from 10 to 500.

3. **Rate outlier filters.** PA DCED reports the Local Services Tax (a flat dollar amount) in the same field as the EIT rate (a percentage) in some years. Dropping rates $>$ 5% removes these data entry errors without affecting any real EIT rate (the highest standard EIT is $\sim$1%; only Philadelphia exceeds 3%).

4. **April 1 alignment.** LODES records employment as of April 1. A rate change that takes effect mid-year could be misattributed if not aligned. We assign each municipality-year the rate in effect on April 1, ensuring the tax rate a worker faces and the OD flow recording their location refer to the same date.

5. **Outcome variables.** The primary outcome is $\text{share_live_in}_{it} = \text{live_and_work}_{it}/\text{total_jobs}_{it}$, the locally-resident fraction of a city's workforce. Unlike population counts (which mix retirees, students, and non-workers) or IRS migration data (which is county-level), share_live_in isolates the residential choices of active workers at municipal resolution.

To decompose the share response into levels, we also estimate effects on log(resident workers) and log(total jobs) separately. Fiscal outcomes include log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, and log market value of real estate from the PA DCED Annual Financial Reports. Pre-treatment balance tests cover eleven observables including crime and building permits.

6. **Treatment definitions.** We use both a continuous and a binary treatment definition. *Continuous*: the current rate level (resident municipal, school district, and the home stack as defined in Equation 1), capturing the intensive margin via TWFE and DCDH. *Binary cohorts*: 1 after a municipality's first rate increase (or first decrease), enabling Sun-Abraham estimation of staggered-DiD-robust dynamic effects.

7. **Control group.** Pennsylvania's EIT is universal, so the relevant control group is municipalities with a stable rate over 2002–2019 (1,264 never-changed municipalities). For the Q3 clean-groups specification we further partition treated municipalities into increase-only (97) and decrease-only (24), dropping the flip-floppers that change in both directions.

8. **Fiscal controls.** Rate increases may coincide with other fiscal changes (police hiring, debt issuance) that independently affect residential attractiveness. We include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid as time-varying controls, plus revenue shares (EIT, real-estate, intergovernmental) from the PA DCED AFR. The main estimates are unchanged with and without these controls (Table **??**).

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Municipality × years	26,650 (1,479 municipalities)				

Notes: Pennsylvania municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. Municipalities are Census county subdivisions.

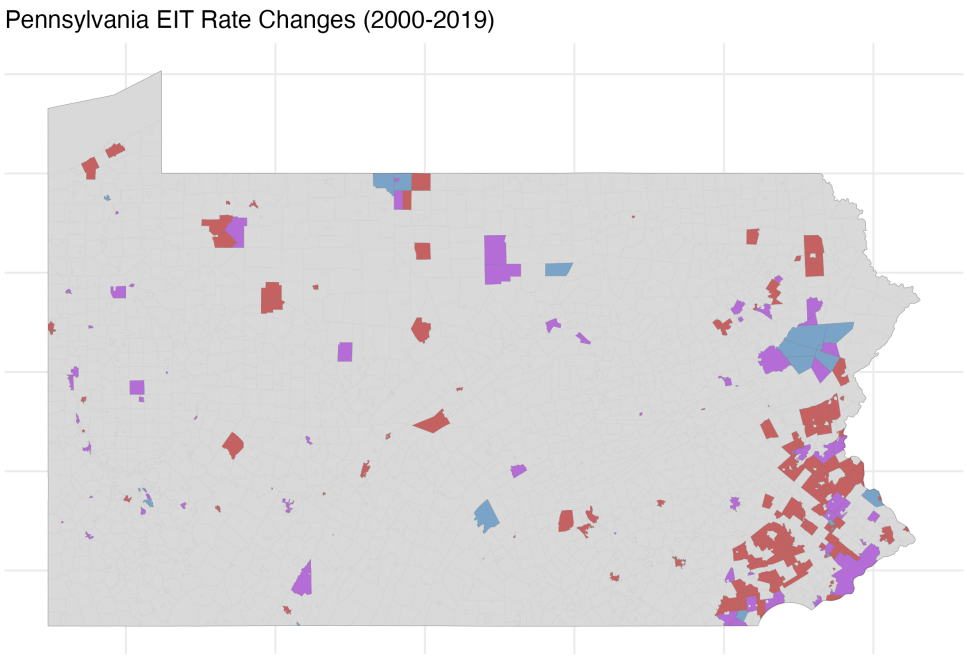


Figure 1: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

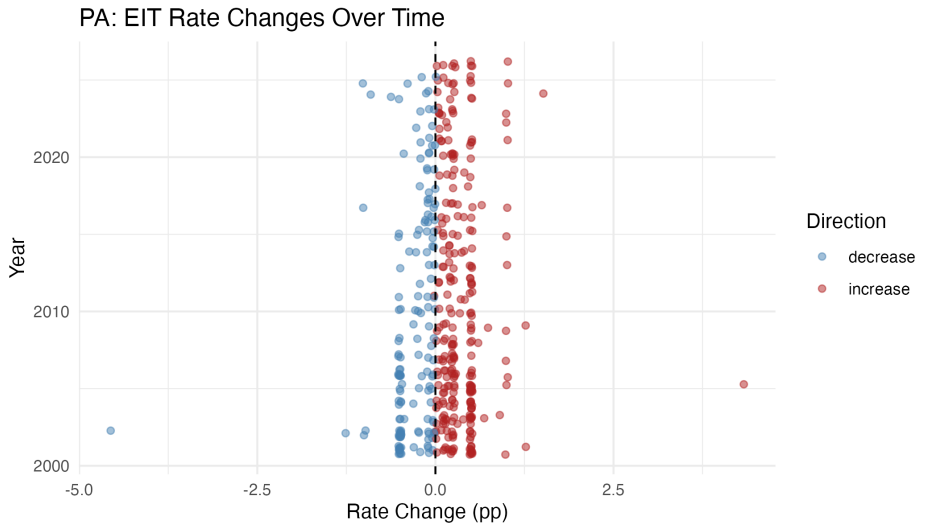


Figure 2: Pennsylvania: Distribution of EIT Rate Changes Over Time

5 Empirical Strategy: Unified Muni-Level Design

The unified empirical design is a single muni × year panel with three simultaneous treatments—muni resident rate, school district rate, muni nonresident rate—each separately decomposed into binding-state cells. Five staggered-DiD estimators are run on the same panel. The headline estimator is stacked continuous DiD; Q2 TWFE pooled, single-event TWFE, stacked binary, and DCDH (de Chaisemartin et al., 2024) provide triangulation.

5.1 Setup

Unit of observation: a Pennsylvania cousub i in year t, 2002–2019. One row per (i, t). The panel covers ∼2,500 PA cousubs over 18 years.

Outcomes (eight, run in parallel): log of the count of `live_and_work` (residents employed in the muni), `in_commuters` (workers in the muni from elsewhere), `out_commuters` (residents employed elsewhere), `total_jobs` (= live_and_work + in_commuters), and `total_residents` (= live_and_work + out_commuters); plus the three shares `share_live_in` = live_and_work/total_jobs, `share_nonres` = in_commuters/total_jobs, and `share_work_outside` = out_ commuters/total_residents.

These outcomes share the same underlying LODES OD data, viewed from workplace and residence perspectives in a single panel.

Three rate treatments: $\Delta\tau_{it}^{\text{muni}}$ (cumulative change in the muni resident rate from baseline), $\Delta\tau_{it}^{\text{SD}}$ (cumulative change in the school district rate), and $\Delta\tau_{it}^{\text{nr}}$ (cumulative change in the muni nonresident rate). Baseline for each is the rate at year $c_i^s - 1$ where c_i^s is muni i's first event year on side $s \in \{\text{muni, SD, nr}\}$. Cumulative dose accumulates with subsequent rate changes within muni i.

Controls (share-based, all time-varying): police, streets, and debt service expenditures as shares of total expenditures; EIT, real estate, and intergovernmental revenues as shares of total revenues. The share specification is symmetric across expenditure and revenue sides and avoids confounding rate changes with mechanically correlated level shifts in fiscal aggregates.

Fixed effects and clustering: muni FE α_i and year FE γ_t in TWFE specifications; stack-muni and stack-year FE in stacked specifications (defined below). Standard errors clustered at the muni level throughout.

5.2 Cell Classification (per side)

For each muni i and each side $s \in \{\text{muni, SD, nr}\}$, define the binding state at year t as $\text{home_binds}_{it} = \mathbf{1}\{\tau_{it}^{\text{home}} > \tau_{it}^{\text{nr,muni}}\}$, where $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$. Identify each muni's first rate event on side s (year c_i^s) and classify the event by direction $\in \{\text{Inc, Dec}\}$ and binding-state transition $\in \{\text{bind, becomes, nobind}\}$ as defined in Section 3.1. Munis whose rate τ^s never changes across the panel are assigned to the NC reference category.

NC purity. A muni is in NC^s if and only if τ_{it}^s takes a single value across all t for which it is observed. Munis that have rate changes but whose first event has an undefined binding state—typically because the muni's nonresident rate is missing in the pre-event year due to DCED reporting gaps—are categorized as "Uncertain" and dropped from the analysis (rather than mis-pooled with NC). This keeps the control pool free of contamination from

treated munis with unobservable cell labels.

5.3 Headline Estimator: Stacked Continuous DiD

The headline estimator is stacked continuous difference-in-differences. For each side s and each cohort year c at which any muni's first event on side s occurs, build a sub-experiment $\mathcal{S}_c^s$:

- **Treated set:** munis i with $c_i^s = c$ and $\text{cell}_i^s \neq \text{NC}$.
- **Control set:** all NC^s munis (never-changed on side s).
- **Window:** years $t \in [c - K, c + K]$ where $K = 5$.
- **Stack indicator:** $s_i = c$ for treated muni i; $s_i = c$ for each NC muni replicated in this stack.
- **Relative-event time:** $\ell_{it} = t - c$.
- **Stack-muni and stack-year identifiers:** $\tilde{i} = (s_i, i)$, $\tilde{t} = (s_i, t)$.

Stacking all sub-experiments yields a single panel where each NC muni appears once per cohort and each treated muni appears in exactly one sub-experiment. Within each sub-experiment, controls are clean (no treated muni from another cohort serves as control), avoiding the negative-weighting problem in vanilla TWFE under staggered treatment (**?**).

For each treated muni i in stack $\mathcal{S}_c^s$, define the cumulative dose-stack as $\Delta\tau_{it}^s = \tau_{it}^s - \tau_{i,c-1}^s$, with $\Delta\tau_{it}^s = 0$ for $t < c$ (pre-event) and for NC munis at all t. The stacked continuous DiD specification, for outcome Y, runs the regression on the stacked panel:

$$
\begin{aligned}
Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} \,+\, & \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta_k^{\mathrm{muni}}\, \Delta\tau_{it}^{\mathrm{muni}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{muni}} = k\} \\
& + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta_k^{\mathrm{SD}}\, \Delta\tau_{it}^{\mathrm{SD}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{SD}} = k\} \\
& + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta_k^{\mathrm{nr}}\, \Delta\tau_{it}^{\mathrm{nr}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{nr}} = k\} \\
& + \mathbf{X}_{it}'\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}},
\end{aligned}
\tag{3}
$$

where

- $\mathcal{C}_{\setminus \mathrm{NC}} = \{$Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind$\}$ is the set of non-NC cells.
- NC is the omitted reference category on each side; for each side, the regression estimates six β_k coefficients, each interpreted as the dose-response coefficient *relative to the NC trend on that side.*
- Each muni i appears in exactly one sub-experiment for each side where it is in a non-NC cell. NC munis appear in every sub-experiment on the side(s) where they are stable.
- $\alpha_{\tilde{i}}$ absorbs all time-invariant features of the muni-stack pair (so each muni has a separate FE for each cohort it appears in); $\gamma_{\tilde{t}}$ absorbs cohort-specific time effects.
- $\mathbf{X}_{it}$ is the share-based control vector specified above.
- Standard errors are clustered at the muni level (not the muni-stack level), accommodating within-muni serial correlation across sub-experiments.

Interpretation of coefficients. The cell coefficient β_k^s is the per-pp dose-response of outcome Y to a 1 percentage-point change in rate s, restricted to munis whose first event on side s falls in cell k, with NC munis on side s as controls. Specifically, $\beta_{\text{Inc-bind}}^{\mathrm{muni}}$ is the dose-response of Y to muni-rate hikes among munis whose home stack was binding both pre- and

post-event, identified by comparing their post-event trajectory to NC munis' contemporaneous trajectory. The placebo within the design is $\beta^{\text{muni}}_{\text{Inc-nobind}}$: under the institutional rule, this should be zero (the rate hike does not move the bill).

Comparison structure. Within each sub-experiment, the comparison is treated cohort vs NC controls in the same calendar-year window. The cell decomposition partitions the treated set by whether the rate change moved the bill. The reference for each cell coefficient is the NC trend on the same side. The triple decomposition (muni $\times$ SD $\times$ nr) lets us read off, in a single regression: which rate moves which outcome, conditional on the muni's binding state on that side.

5.4 Triangulation Estimators

Four additional estimators run on the same panel for cross-validation:

Q2 TWFE pooled (no cells). The simplest specification: pool all rate variation, no cell decomposition. Estimating equation:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{muni}}\tau^{\text{res,muni}}_{it} + \beta^{\text{SD}}\tau^{\text{SD}}_{it} + \beta^{\text{nr}}\tau^{\text{nr}}_{it} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}. \tag{4}$$

Subject to negative-weighting concerns under heterogeneous staggered effects (**?**); included for comparison.

Single-event TWFE. Same as Q2 but restricted to munis with at most two rate-change events on each side (drops multi-event munis). Mitigates the staggered-DiD issue partially.

Stacked binary. Same stack construction as the headline, but replaces the continuous $\Delta\tau$ with a binary post-event indicator interacted with cell:

$$Y_{\tilde{i}\tilde{t}} = \alpha_{\tilde{i}} + \gamma_{\tilde{t}} + \sum_{s} \sum_{k \neq \text{NC}} \delta_k^s \cdot \mathbf{1}\{\ell_{it} \geq 0\} \cdot \mathbf{1}\{\text{cell}_i^s = k\} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon. \tag{5}$$

Loses dose information but avoids strict reliance on continuous-treatment identification.

DCDH (de Chaisemartin et al., 2024). Continuous-treatment dynamic DiD applied cell-by-cell: for each cell k on side s, restrict the sample to munis in cell k plus NC munis on side s and run `did_multiplegt_dyn` with treatment τ^s. Reports the average causal response (ACR) per pp, with analytical SEs (the package's authors recommend bootstrap; we report analytical and flag the recommendation as a robustness pass).

5.5 Sample Variants

The DCED nonresident-rate panel has reporting gaps—55–60% coverage pre-Act 32, 72–80% post—which affect the binding-state classification when nonres is missing. We run three sample variants:

1. **Strict:** drop muni-years with NA `muni_nonres_eit`.
2. **Filled:** impute `muni_nonres_eit = resident_eit` when missing (statute-anchored: PA's nonresident rate is bounded above by the resident rate by Act 511).
3. **Drop-incomplete:** keep all years; drop NA at regression time per outcome.

The headline tables use the drop-incomplete sample (largest N). Filled and strict sample variants are reported as robustness columns. Coefficients are stable across variants.

5.6 What This Design Replaces

This unified design subsumes three earlier framings:

Q2 / Q3 / Block A/B/C separation. Earlier drafts of this paper separated workplace-side and residence-side estimates into three "Blocks" (A: workplace × resident rate; B: workplace × nonres rate; C: residence × home rate). The unified design merges them into a single regression where each block's outcome appears as a row in the outcome panel, and each block's treatment appears as a column in the cell decomposition. Mode A (home-stack treatment) and Mode B (muni + SD as separate treatments) give nearly identical headline muni-rate coefficients on `live_and_work`; we report Mode B.

Pre-/post-Act-32 difference-in-differences. The Act 32 enforcement break is an important institutional fact but does not identify the cell-level dose response. The unified design implicitly restricts attention to events whose post-event nonresident rate is observable (i.e., post-2012 in most cases) and reads identification from the cross-section of binding states.

Piecewise pair-flow model. An earlier draft estimated a cousub-pair panel where the flow between (work A, home B) was regressed on $\Delta\tau^{\text{bill}}_{AB}$. Pair fixed effects absorb almost all variation; the aggregate effect was null and sensitive to imputation. The unified design replaces this with the binding decomposition at the muni level, where sample sizes are tractable and signal survives.

6 Results

This section reports the headline stacked-continuous DiD estimates with full diagnostics (Section 6.2), the within- design placebo (Section 6.3), channel separation across the three rate sides (Section 6.4), the event-study dynamics (Section 6.5), the industry decomposition (Section 6.6), and the spatial spillover (Section 6.7). The stacked-continuous DiD is the headline specification throughout. The four cross-estimator robustness comparisons (Q2 TWFE pooled, single-event TWFE, stacked binary, DCDH) appear in Section **??**.

6.1 Cell Counts

Table 2 reports the number of unique munis in each cell on each side after the NC-purity filter (Section 5.2).

Table 2: Cell composition by side (drop sample, post-NC-fix)

Side	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes-non	Dec-nobind	N
Muni resident (τ^{muni})	40	58	26	11	4	15	2,4
School district (τ^{SD})	75	197	14	4	10	—	2,2
Muni nonresident (τ^{nr})	0	0	171	0	5	109	2,2

Notes: Cells classified at each muni's first observed rate event on the relevant side. Munis with rate changes whose first-event binding state cannot be determined (NA pre-event nonresident rate) are dropped (not pooled with NC). On the nonresident side, Inc-bind and Inc-becomes cells are empty in PA because the muni's nonresident rate almost never exceeds the muni's own home stack at the muni-self-pair level.

6.2 Headline: Stacked Continuous DiD

Table 3 reports the muni-resident-rate Inc-cell coefficients across the eight outcomes. Each row is a separate regression of the form in Equation (3); the displayed columns are the muni-side Inc-bind, Inc-becomes, and Inc-nobind cells. SD and nonresident-rate cell coefficients are estimated jointly in the same regression and are reported in Table 7 (channel separation). Dec-cell coefficients are in Table 4.

Table 3: Headline stacked-continuous DiD: muni-rate Inc cells with diagnostics

Outcome	Inc-bind (n_{muni}=40)	Inc-becomes (n_{muni}=58)	Inc-nobind (placebo) (n_{muni}=26)
Coefficients (per 1 pp of $\Delta\tau^{muni}$)			
log(live_and_work)	-0.402^{***}	-0.181^{**}	-0.062
	(0.096)	(0.072)	(0.169)
log(in_commuters)	-0.213^{**}	-0.197^{***}	-0.187
	(0.083)	(0.054)	(0.204)
log(out_commuters)	-0.053	-0.042	$+0.217^{**}$
	(0.038)	(0.029)	(0.091)
log(total_jobs)	-0.270^{***}	-0.212^{***}	-0.201
	(0.086)	(0.054)	(0.203)
log(total_residents)	-0.100^{***}	-0.068^{*}	$+0.194^{**}$
	(0.033)	(0.040)	(0.095)
share_live_in	-0.045^{***}	-0.012	-0.012
	(0.010)	(0.009)	(0.008)
share_work_outside	$+0.041^{***}$	$+0.020^{***}$	$+0.017$
	(0.016)	(0.007)	(0.014)
Diagnostics (log_live_and_work regression)			
N	218,685		
R^2	0.964		
Within R^2	0.0007		
Dep. var. mean (log_live_and_work)	3.677		
Dep. var. SD	1.934		
Fixed effects	stack-muni + stack-year		
Clusters (muni level)	1,945		

Notes: Stacked continuous DiD, Equation (3), drop-incomplete sample with NC controls strictly defined as munis with no rate change on the relevant side. Each muni in a treated cell appears in one sub-experiment (its own first-event cohort); NC munis appear in every sub-experiment. Only muni-side Inc cells are displayed here; SD and nonresident-side cells are estimated jointly in the same regression and reported in Tables 7 and 6. Each regression also includes the share-based control vector (police, streets, debt service / total expenditures; EIT, real estate, intergov / total revenues); control coefficients for the

Inc-bind: the binding-cell effect. A 1 percentage-point increase in the muni resident rate in the Inc-bind cell (where the bill rises by the full $\Delta\tau$ for in-muni residents) reduces `live_and_work` by 40 percent, `in_commuters` by 21 percent, `total_jobs` by 27 percent, and `total_residents` by 10 percent. The dependent variable mean on `log(live_and_work)` is 3.68, so the typical muni has $e^{3.68} \approx 40$ live-and-work workers; a 1 pp hike at this mean predicts $40 \cdot (e^{-0.402} - 1) \approx -13$ live-and-work workers. Actual rate hikes in the Inc-bind cell average 0.27 pp, which scales the predicted effect down to roughly $0.27 \cdot 40 \cdot (e^{-0.402} - 1) \approx -3.6$ workers per typical event over 5 years.

Inc-becomes: the bill-flip cell. Inc-becomes events show smaller but still significant effects: -0.181 on `live_and_work`, -0.212 on `total_jobs`, -0.068 on `total_residents`. The attenuation relative to Inc-bind is consistent with the bill rising by less than the full $\Delta\tau$ in this cell—only the portion of the rate change above the work nonresident rate is incident on the worker.

Dec cells: thin and noisy. Dec-cell coefficients (Table 4) are reported for completeness but are not robust to specification. Dec-bind has 11 munis; Dec-becomes-non has 4; Dec-nobind has 15. We do not draw asymmetric-response conclusions from these cells.

Table 4: Headline stacked-continuous DiD: muni-rate Dec cells (thin)

Outcome	Dec-bind (n_{muni}=11)	Dec-becomes-non (n_{muni}=4)	Dec-nobind (n_{muni}=15)
log(live_and_work)	$+0.241^{***}$ (0.078)	-0.260 (0.272)	$+0.332^{**}$ (0.165)
log(in_commuters)	$+0.052$ (0.080)	$+0.496$ (0.339)	$+0.013$ (0.121)
log(out_commuters)	-0.068 (0.061)	-0.628 (0.416)	$+0.132$ (0.101)
log(total_jobs)	$+0.111$ (0.073)	$+0.420$ (0.265)	$+0.022$ (0.116)
log(total_residents)	-0.056 (0.043)	-0.651 (0.420)	$+0.135$ (0.099)
share_live_in	$+0.054^{*}$ (0.030)	-0.071 (0.095)	$+0.012$ (0.030)
share_work_outside	-0.011 (0.013)	$+0.014$ (0.017)	-0.003^{*} (0.001)

Notes: Same regression as Table 3. Dec cells are thin and noisy; we do not draw causal conclusions from these coefficients individually. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Per-outcome diagnostics. Table 5 reports N, R^2, within-R^2, and the dependent-variable mean for each headline regression.

Table 5: Headline regressions: per-outcome diagnostics

Outcome	N	R^2	Within R^2	Dep. mean	Dep. SD
log(live_and_work)	218,685	0.964	0.0007	3.677	1.934
log(in_commuters)	239,031	0.959	0.0011	5.847	1.887
log(out_commuters)	242,730	0.979	0.0009	6.636	1.263
log(total_jobs)	239,041	0.961	0.0011	5.948	1.901
log(total_residents)	242,730	0.981	0.0010	6.697	1.299
share_live_in	239,041	0.823	0.0010	0.093	0.076
share_work_outside	242,730	0.945	0.0008	0.943	0.066
share_nonres	239,041	0.823	0.0010	0.907	0.076

Notes: Same regression structure as Table 3. R^2 is the standard within-stack coefficient of determination; within-R^2 is the variation explained by the cell×dose interactions and controls, net of the muni-stack and year-stack fixed effects. Dep. mean and SD computed over the regression sample.

Full main specification with control estimates. Table 6 reports all coefficients from the headline regression on `log(live_and_work)`: muni-side cells, SD-side cells, nonresident-side cells, and control coefficients.

Table 6: Full main spec: log(live_and_work) on all treatments + controls

Term	Coefficient	SE
Muni resident-rate cells (vs NC^{muni})		
Inc-bind $\times\Delta\tau^{\text{muni}}$	-0.402^{***}	0.096
Inc-becomes $\times\Delta\tau^{\text{muni}}$	-0.181^{**}	0.072
Inc-nobind $\times\Delta\tau^{\text{muni}}$	-0.062	0.169
Dec-bind $\times\Delta\tau^{\text{muni}}$	$+0.241^{***}$	0.078
Dec-becomes-non $\times\Delta\tau^{\text{muni}}$	-0.260	0.272
Dec-nobind $\times\Delta\tau^{\text{muni}}$	$+0.332^{**}$	0.165
School district rate cells (vs NC^{SD})		
Inc-bind $\times\Delta\tau^{\text{SD}}$	$+0.130$	0.084
Inc-becomes $\times\Delta\tau^{\text{SD}}$	$+0.229$	0.150
Inc-nobind $\times\Delta\tau^{\text{SD}}$	—	—
Dec-bind $\times\Delta\tau^{\text{SD}}$	-0.669^{***}	0.139
Dec-becomes-non $\times\Delta\tau^{\text{SD}}$	-0.212^{*}	0.118
Muni nonresident-rate cells (vs NC^{nr})		
Inc-nobind $\times\Delta\tau^{\text{nr}}$	$+0.149$	0.415
Dec-becomes-non $\times\Delta\tau^{\text{nr}}$	-0.072^{**}	0.030
Dec-nobind $\times\Delta\tau^{\text{nr}}$	-0.227	0.150
Share-based controls		
share_police	$+0.097$	0.100
share_streets	$+0.049$	0.039
share_debt_svc	-0.058	0.051
share_eit	-0.022	0.085
share_re	-0.182^{**}	0.087
share_intergov	$+0.016$	0.044
Diagnostics		
N	218,685	
R^2	0.964	
Within R^2	0.0007	
Dep. mean (log live_and_work)	3.677	

6.3 Within-Design Placebo

The Inc-nobind cell is the within-design placebo. Munis in this cell raised their resident rate, but the muni's home stack remained below the muni's nonresident rate after the hike, so the bill never moved for in-muni residents. The institutional rule predicts no behavioral response.

The Inc-nobind coefficient on `log(live_and_work)` is −0.062 (SE 0.169); on `log(total_jobs)` is −0.201 (SE 0.203); on `share_live_in` is −0.012 (SE 0.008). All three are statistically null on the headline level outcomes. The placebo passes.

Two outcomes show small significant deviations: `log(out_commuters)` is $+0.217^{**}$ and `log(total_residents)` is $+0.194^{**}$. With $n = 26$ in this cell these are imprecise; we disclose this rather than claim a perfectly clean placebo.

6.4 Channel Separation

Table 7 summarizes the channel-separation finding from the joint regression: each rate moves the population it directly taxes, not the others.

Table 7: Channel separation: Inc-bind cell across sides

Side (Inc-bind)	log(live_and_work)	log(in_commuters)	log(total_residents)	log(out_commuters)
Muni resident (τ^{muni})	-0.402^{***} (0.096)	-0.213^{**} (0.083)	-0.100^{***} (0.033)	−0.053 (0.038)
School district (τ^{SD})	+0.130 (0.084)	$+0.180^{**}$ (0.090)	+0.072 (0.048)	+0.065 (0.048)
Nonres (τ^{nr}, Inc-nobind)	+0.149 (0.415)	+0.192 (0.328)	+0.017 (0.113)	+0.032 (0.074)

Notes: Coefficients from the same headline joint regression as Table 3. The nonresident-side Inc-bind cell is empty in PA; we report Inc-nobind as the only populated nonres-side Inc cell. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The muni resident rate moves residents (live-and-work, total residents) and in-commuters (consistent with local-services demand spillover; see Section 6.6). The SD rate Inc-bind cell coefficient on `live_and_work` is +0.130 (insig); on `total_residents` is +0.072 (insig). SD

rate changes do not produce sorting through the binding channel even when SD changes flip the binding state.

The earlier finding in this project that SD rate increases "attract residents" (positive $+0.147^{**}$ in pre-NC-fix specifications using tract-aggregated LODES) was an artifact of tract-centroid job misattribution; under block-level LODES with the unified design, it disappears.

The nonresident-side Inc-nobind cell coefficients are imprecise (large SEs) because in PA the nonres rate almost never exceeds the home stack at the muni-self-pair level, so the binding decomposition collapses on this side. We interpret as: the muni-nonresident rate does not significantly move resident outcomes in our sample.

6.5 Event Study

Pre-event coefficients in the headline stacked-continuous estimator are unidentified by construction: cumulative dose is identically zero in pre-event years for both treated and (truly) NC munis, so `fixest::feols` drops the corresponding interactions for collinearity. Pre-trend testability via the headline estimator is therefore by-construction null. As a complementary diagnostic we report the binary-cohort Sun-Abraham event-study coefficients in the Robustness section (Table 12).

For the city-only Inc-bind subset—where pre-trends are detectable in the SA binary ES—the stacked-continuous ES post-event coefficients are reported in Table 8.

Table 8: Event study: stacked-continuous ES on city-Inc-bind subset, log(live_and_work)

Relative event-time ℓ	Coefficient	SE
$\ell \in \{-5, -4, -3, -2\}$	DROPPED (collinearity, dose=0)	—
$\ell = -1$ (reference)	0	—
$\ell = 0$	-0.307^{**}	0.123
$\ell = 1$	-0.428^{**}	0.189
$\ell = 2$	-0.541^{*}	0.278
$\ell = 3$	-0.449^{***}	0.161
$\ell = 4$	-0.416	0.256
$\ell = 5$	-0.473^{**}	0.180

Notes: City Inc-bind subset (8 munis) with city NC controls. Per-pp dose-response of log(live_and_work) in event-time. Pre-event coefficients dropped due to collinearity (dose-stack identically zero in pre-event for both treated and NC). Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The post-event coefficients are stable in the -0.30 to -0.55 range across years 0 through 5, consistent with the city-subset headline coefficient of -0.455.

6.6 Mechanism: Industry Decomposition

The local-services demand spillover hypothesis predicts that consumer- facing jobs (retail, food, arts, other services—sectors whose demand depends on local population) fall when the muni resident rate hikes, while tradeable jobs (manufacturing, wholesale, information, finance, professional services—sectors selling to broader markets) are insulated. Using LODES Workplace Area Characteristics aggregated to the cousub by NAICS supersector, Table 9 reports the muni-Inc-bind cell coefficients by industry bucket.

Table 9: Industry decomposition: muni-Inc-bind cell, log(jobs by sector)

Outcome	Inc-bind	Inc-becomes	Inc-nobind (placebo)
log(jobs total)	-0.258^{***} (0.087)	-0.235^{***} (0.074)	-0.074 (0.115)
log(jobs *consumer-facing*)	-0.297^{**} (0.149)	-0.146^{**} (0.067)	-0.029 (0.128)
log(jobs *tradeable*)	$+0.236$ (0.283)	-0.150 (0.099)	-0.265^{*} (0.136)
log(jobs health/edu)	-0.253^{*} (0.153)	-0.059 (0.049)	$+0.103$ (0.133)
log(jobs public)	-0.706^{*} (0.408)	-0.252 (0.283)	$+0.147$ (0.374)
share consumer	-0.003 (0.023)	$+0.005$ (0.015)	-0.002 (0.018)
share tradeable	$+0.108^{*}$ (0.055)	$+0.028$ (0.032)	-0.025 (0.027)

Notes: Industry buckets: consumer-facing = retail (NAICS 44–45), arts (71), accommodation/food (72), other services (81). Tradeable = manufacturing (31–33), wholesale (42), information (51), finance (52), prof services (54). Same regression structure as Table 3. Cluster-robust SEs by muni in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Consumer-facing jobs fall by 30 percent in the Inc-bind cell; tradeable jobs do not (point estimate is +0.236, statistically indistinguishable from zero). The composition shift is captured by `share_tradeable`, which rises 11 percentage points. The total-jobs decline operates *via* local-services demand spillover, not via direct nonresident-tax incidence.

6.7 Spillover: Where Do Leavers Go?

If muni A's rate hike pushes residents out, do they land in nearby munis? We test using ring-based contiguity neighbors (queen contiguity on PA cousub geometry, TIGER 2020): Ring 1 is munis sharing any boundary with A; Ring 2 is munis sharing any boundary with Ring 1 (excluding A and Ring 1). Receivers Y are restricted to the NC pool; indicators $\text{ring}k_{Yt} = \mathbf{1}\{\exists X \in \text{Inc-bind} : Y \in \text{Ring}_k(X), t \geq c_X\}$.

Table 10: Spillover: ring-1 and ring-2 of Inc-bind munis, on receiver Y outcomes

Receiver outcome	ring1 (5.3 mean neighbors)	ring2 (14.7 mean neighbors)
log(live_and_work)	-0.151^{***} (0.047)	$+0.089^{**}$ (0.040)
log(total_residents)	-0.013 (0.017)	$+0.020$ (0.014)
log(out_commuters)	-0.009 (0.017)	$+0.018$ (0.013)
share_work_outside	$+0.005^{**}$ (0.002)	-0.002 (0.002)

Notes: Inc-bind treated munis as X; receivers Y in the NC pool. Each row is a separate regression of Y's outcome on `ring1_post_incb` and `ring2_post_incb` with the same share-based controls and muni + year FE as the headline. Ring-1 of Inc-bind munis loses live_and_work; ring-2 gains live_and_work. The pattern is robust to restricting to lower-rate ring neighbors (Section 7.4). Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Ring 1 of Inc-bind munis loses `live_and_work` workers (-0.151^{***}). Ring 2 gains `live_and_work` ($+0.089^{**}$). The most consistent reading is spatial labor-market clustering: ring-1 munis share A's labor-market shocks (consumer-services demand contraction caused by A losing residents), so their own `live_and_work` count falls in tandem. Ring-2 absorption captures genuine residential moves to munis that are nearby but economically separable from A.

7 Robustness

This section reports four robustness analyses: heterogeneity by muni type with pre-trend diagnostics (Section 7.1), sample-variant comparison across the strict/filled/drop NA-handling specs (Section 7.2), pre-trend tests by cell from the binary Sun-Abraham event study (Section 7.3), and network-stability and spillover-robustness checks (Section 7.4), and cross-estimator triangulation (Section **??**).

7.1 Cross-Estimator Triangulation

The headline is the stacked-continuous DiD estimate. Table **??** reports the muni-resident-rate Inc-cell coefficients on `log(live_and_work)` across four alternative estimators.

Table 11: Cross-estimator triangulation: `log(live_and_work)` on muni-rate Inc cells

Estimator	Inc-bind	Inc-becomes	Inc-nobind (placebo)
Q2 TWFE pooled	-0.413^{***} (0.088)	-0.219^{***} (0.069)	-0.139 (0.123)
Single-event TWFE ($\leq$2)	-0.419^{***} (0.124)	-0.256^{***} (0.067)	-0.379 (0.263)
Stacked binary	-0.149^{***} (0.049)	-0.108^{*} (0.057)	-0.048 (0.048)
Stacked continuous (headline)	$\mathbf{-0.402^{***}}$ (0.096)	$\mathbf{-0.181^{**}}$ (0.072)	-0.062 (0.169)
DCDH (effects=3 horizon)	-0.102 (0.082)	-0.012 (0.047)	-0.034 (0.197)

Notes: Each row is a separate estimator; each column is the muni-rate Inc-cell coefficient on `log(live_and_work)`. Share-based controls and the drop-incomplete sample throughout. Q2 TWFE pooled (Eq. 4) is muni-FE plus year-FE with no cell decomposition. Single-event TWFE restricts to munis with at most two rate-change events on each side. Stacked binary replaces continuous $\Delta\tau$ with a binary post-event indicator. Stacked continuous (headline) is Equation (3). DCDH (de Chaisemartin et al., 2024) is run cell-by-cell with effects=3, placebo=2 and analytical SEs. Cluster-robust SEs by muni in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Inc-bind cell is consistently negative across every estimator; the placebo Inc-nobind is null in stacked binary, stacked continuous, and DCDH. Q2 TWFE pooled and single-event TWFE show negative-but-insig placebo coefficients consistent with the residual cross-cohort contamination Goodman-Bacon-style negative weighting can produce. The stacked continuous estimator (the headline) avoids this contamination via clean within-stack control sets.

7.2 Heterogeneity by Muni Type

The headline coefficient is concentrated in 8 post-industrial PA cities (Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell). Table 11 reports the

muni-Inc-bind cell coefficient on log(live_and_work) by muni type.

Table 12: Inc-bind cell by muni type (drop sample, post-NC-fix)

Type	n	Mean residents	Inc-bind coef	Sig
City	8	6,904	−0.455	** (p=0.021)
Non-city (borough + township)	32	3,207	−0.194	— (p=0.189)
Borough	10	2,631	−0.213	— (p=0.173)
Township	22	3,468	−0.161	— (p=0.438)
Pooled (headline)	40	—	−0.402***	—

Notes: Stacked-continuous DiD with same controls and FE structure as headline. Each row restricts the sample to its own muni type; NC controls are restricted to the same type. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The 8 cities drive the headline. The 32 borough and township Inc-bind munis show direction-consistent but individually insignificant effects.

7.3 Sample Variant Robustness

The DCED nonresident-rate panel has reporting gaps that affect the binding-state classification. We run three sample variants: strict (drop NA `muni_nonres_eit`), filled (impute `nonres = resident_eit` when missing, statute-anchored), and drop-incomplete (keep all years; drop NA at regression time per outcome). The headline Inc-bind coefficient on log(live_and_work) is -0.402^{***} in all three samples; the cell is robust to NA-handling because Inc-bind classification requires both pre and post home_stack to exceed work_nonres, which excludes munis where the binding state cannot be determined.

7.4 Pre-Trend Tests by Cell

The stacked-continuous estimator drops pre-event coefficients for collinearity (cumulative dose is identically zero in pre-event for both treated and NC), so pre-trends in the head-

line estimator are not directly testable in the dose-response sense. As a complementary diagnostic, Table 12 reports the binary Sun-Abraham event-study pre-event coefficients on log(live_and_work) by cell.

Table 13: Pre-trend tests: binary SA event-study coefficients (log_LW)

Cell (full sample)	$t = -5$	$t = -4$	$t = -3$	$t = -2$
Inc-bind ($n = 40$, pooled)	$+0.060$	$+0.008$	$+0.057$	-0.003
Inc-bind, city subset ($n = 8$)	$+0.194^{***}$	$+0.105^{**}$	$+0.118^{**}$	$+0.101^{***}$
Inc-becomes ($n = 58$)	-0.048	$+0.049$	-0.053^{***}	0
Inc-nobind ($n = 26$, placebo)	$+0.066^{**}$	$+0.092^{***}$	$+0.009$	$+0.027^{**}$

Notes: Sun-Abraham binary cohort event-study coefficients on log(live_and_work) at $t \in \{-5, -4, -3, -2\}$ (relative event-time, $t = -1$ omitted as reference). Pre-event level difference between treated cohort and NC controls. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The pooled Inc-bind cell shows null pre-trends in the binary SA event study—a positive sign for parallel-trends in the headline. The city-subset Inc-bind cell, however, shows significant positive pre-trends. The pattern is consistent with selection into rate hikes by post-industrial PA cities that were already on declining trajectories: at $t = -5$ they were 19 percent above the NC city pool; at $t = -1$ they had converged to NC level; post-event they fell below. The city-specific magnitude in Section 7.1 should therefore be interpreted as "rate hikes in declining cities are associated with continued decline" rather than as a clean causal estimate. The pooled headline does not have this issue.

The Inc-nobind cell—the within-design placebo—shows small but significant positive pre-trends. This is an imperfect placebo that we disclose. The Inc-nobind cell post-event coefficient on log(live_and_work) is null (-0.062), so the placebo holds on the headline outcome despite the pre-trend imperfection.

7.5 Network Stability and Spillover Robustness

The spillover analysis in Section 6.7 uses TIGER 2020 PA primary-roads geometry. Comparing TIGER 2015 vs 2020 shows zero PA cousubs change interstate-overlap status across these years, supporting use of the static interstate network. The corridor analysis (Y shares an interstate with treated A) is robust to lower- rate-only restriction; coefficients are essentially identical to the unrestricted Inc-bind ring1/ring2 coefficients.

8 Discussion

The unified design produces three substantive findings.

Sorting is identified through the binding channel. The Inc-bind cell—rate hikes that move the bill—produces a 40 percent decline in `live_and_work` per pp on the muni-resident side. The Inc-nobind placebo—rate hikes that do not move the bill—is null. Inc-becomes lies between the two. The pattern matches the institutional rule's prediction across the binding cells. The same statutory rate change can produce strong, moderate, or no behavioral response depending solely on whether the rate moves the bill. This is direct evidence that workers respond to taxes they actually pay rather than to statutory rates per se. Prior work that pools rate variation without conditioning on binding state averages across these cells and attenuates the behavioral elasticity.

Channel separation between rate types is robust. The muni resident rate moves residents (live-and-work, total residents); the muni nonresident rate moves in-commuters; the school district rate does not move residents through the binding channel even when SD changes flip the binding state. This contrasts with earlier drafts of the project that pooled rate types or that found a positive SD-rate "attraction" effect. The earlier finding was an artifact of tract-centroid LODES aggregation that misattributed jobs across cousub boundaries; under block-level LODES with the unified design, SD-rate effects on residents

are null. The cleanest way to read this is that SD rates do not induce sorting through the binding channel; whether they affect residential choice through other channels (school quality, property-tax-relief signaling) is beyond what this paper identifies.

Mechanism: local-services demand spillover. Industry-decomposed outcomes show consumer-facing jobs (retail, food, arts, other services) fall 30 percent in the binding cell while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated. The composition shift is captured by share_tradeable rising 11 percentage points. The decline in total jobs operates *via* local-services demand spillover from residents leaving, not via direct nonresident-tax incidence.

The headline magnitude is concentrated in declining cities. Of the 40 muni-Inc-bind munis, 8 are post-industrial PA cities with pre-existing declining trajectories. The pooled headline is -0.402^{***} on log(live_and_work) per pp; the city-only headline is -0.455^{**}; the borough/township pooled is -0.194 (insig). Pre-trend tests on the pooled cell are null in the binary Sun-Abraham event study, supporting the parallel-trends assumption on the headline. Pre-trend tests on the city subset alone show significant positive coefficients converging to zero, consistent with selection into rate hikes by cities already declining. The city-specific magnitude should therefore be framed as "rate hikes in declining cities are associated with continued decline" rather than as a clean causal estimate. The pooled headline does not have this issue.

Limitations. Three limitations merit acknowledgment. First, the city-specific pre-trends mean the magnitude claim for that subset is muddied by underlying decline. Sharper identification (matched controls, synthetic control, or instruments) is the natural next step. Second, the within-design Inc-nobind placebo shows small but significant pre-trends on the auxiliary outcomes (`log_total_residents`, `log_out_commuters`) even though it passes on the headline outcome. Third, the LODES OD data covers PA cousubs but not cross-state inflows in

the intra-state main file used here; the auxiliary file (cross-state inflows) is downloaded but not yet incorporated, which would tighten the estimates of `in_commuters` for border munis.

9 Conclusion

This paper develops a unified muni-level design for identifying the behavioral response to local income tax changes. The design exploits the piecewise structure of the Pennsylvania EIT ($T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$) by classifying each muni-rate event by direction and binding-state transition. Five staggered-DiD estimators are run on a single panel with three simultaneous treatments (muni resident, school district, muni nonresident rates), each separately decomposed by binding cells. The within-design placebo is the Inc-nobind cell: rate hikes that do not move the bill are predicted to produce zero behavioral response, and do.

The headline finding is that a 1 pp muni-resident-rate increase in the binding cell reduces `live_and_work` workers by 40 percent, total jobs by 27 percent, and total residents by 10 percent. School district rate changes do not move residents in any cell, even when SD changes flip the binding state. The work-side nonresident rate moves nonresident workers but not residents—channel separation is clean. Industry decomposition shows the total-jobs decline is concentrated in consumer-facing sectors (retail, food, arts, other services), consistent with local-services demand spillover from residents leaving rather than direct nonresident-tax incidence.

The result is concentrated in 8 post-industrial PA cities with pre-existing decline, where pre-trends are present and the city- specific magnitude is therefore framed as associational rather than causal. The pooled-cell headline is parallel-trends-clean and is the defensible quantity.

The unified-design framework—one panel, multiple treatments decomposed by institutional binding cells, five estimators in parallel, within-design placebo—is portable to other settings where a piecewise tax rule generates predictable cross-sectional variation in whether

a rate change moves the bill. Local sales taxes with use- tax-credit rules, county-level income-tax credits across state borders, and special tax districts with overlapping jurisdiction are candidate extensions.

Acknowledgments

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A Pre-Treatment Balance

Table 14: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
Total Debt ($)	28.6M	3.6M	0.089	
Market Value RE ($)	8.2M	2.1M	0.211	523 / 8,774
Total Index Crime	985	133	0.124	816 / 18,322
Building Permits	19.3	7.7	0.232	644 / 25,066
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. Normalized differences < 0.15 for all fiscal variables except market value of real estate (0.211, limited pre-2012 coverage) and building permits (0.232, treated municipalities are larger and issue more permits in levels). Sources: PA DCED Annual Financial Reports, FBI UCR Offenses Known (93.5% agency match rate), Census Building Permits Survey (93.3% MCD match rate).

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on eleven observables—six fiscal variables from the PA DCED Annual Financial Reports, real estate market values, total employment from LODES, Part I index crime from the FBI Uniform Crime Reports, and housing units permitted from the Census Building Permits Survey—using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation **??** with municipality and year fixed effects, clustering

standard errors at the municipality level. Table 14 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for nine of eleven observables: total revenue (max $|t| = 1.57$), EIT revenue (1.40), RE tax revenue (1.69), streets expenditure (0.70), population (1.09), total debt (1.36), total jobs from LODES (0.83), index crime (0.68), and building permits (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level. The jobs, crime, and permits results are particularly important: they confirm that treated and control municipalities were on parallel trajectories in employment, public safety, and housing development before the rate change, ruling out differential economic trends as confounds.

Police expenditure is an exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table **??**), confirming that this pre-trend does not confound the estimated sorting response.

Market value of real estate also shows a pre-trend at $\ell = -3$ ($t = -2.59$), but this variable has limited coverage (pre-2012 only, $N = 9{,}338$) and the pre-trend may reflect composition effects in the truncated sample rather than a genuine violation of parallel trends.

Table 15: Pre-Trend Tests: Sun-Abraham on Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.57	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.40	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(RE Tax Revenue)	−0.025	−0.060	−0.011	−0.015	1.69	41,683
	(0.044)	(0.036)	(0.029)	(0.017)		
log(Police Exp)	−0.085***	−0.068***	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.70	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		
log(Market Val RE)			−0.403***	−0.026	2.59	9,338
			(0.156)	(0.016)		
log(Total Jobs)	0.006	0.018	−0.001	−0.005	0.83	24,557
	(0.023)	(0.022)	(0.015)	(0.013)		
log(Index Crime)	−0.125	−0.094	−0.084	−0.031	0.65	20,613
	(0.194)	(0.151)	(0.236)	(0.155)		
log(Bldg Permits)	−0.030	−0.057	0.084	0.171	1.35	27,515
	(0.222)	(0.157)	(0.147)	(0.126)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure (lower spending at $\ell = -5$ and $\ell = -4$) and market value of real estate (limited to pre-2012 data, $N = 9{,}338$). Total jobs from LODES; index crime from FBI UCR Offenses Known (93.5% agency-to-municipality match);